



Healthcare Cost & Patient Outcomes Analysis

Hospital Operations Intelligence Report · Power BI · Excel Power Query · DAX

500 PATIENTS

₹4.3M REVENUE

32% INSURED

5.8 DAYS AVG STAY

5 CITIES

Executive Summary

Analysis of 500 patients across 5 Indian cities (Jan–Oct 2024) identified four critical operational and financial issues.

Insurance Gap

Uninsured pay **82% more** — ₹10,059 vs ₹5,540 avg bill.
High financial equity risk.

Pediatrics Cost

Pediatrics costs **6× Orthopedics** — ₹25,000 vs ₹4,000.
Budget allocation concern.

Doctor Workload

Dr. Madhu handles **45% of patients** — 224 of 500.
Operational single point of failure.

Long Stays

64% of patients stay >4 days (320 of 500). Bed capacity pressure.

Insurance Coverage Gap

The 82% Cost Gap

Only 32.2% of patients carry insurance, leaving 339 exposed to full billing rates. Uninsured patients pay nearly double for the same treatment — a significant welfare concern.

 Prioritize enrollment in Pediatrics & General — highest absolute gaps (~108% and ~113%). Partner with Ayushman Bharat.

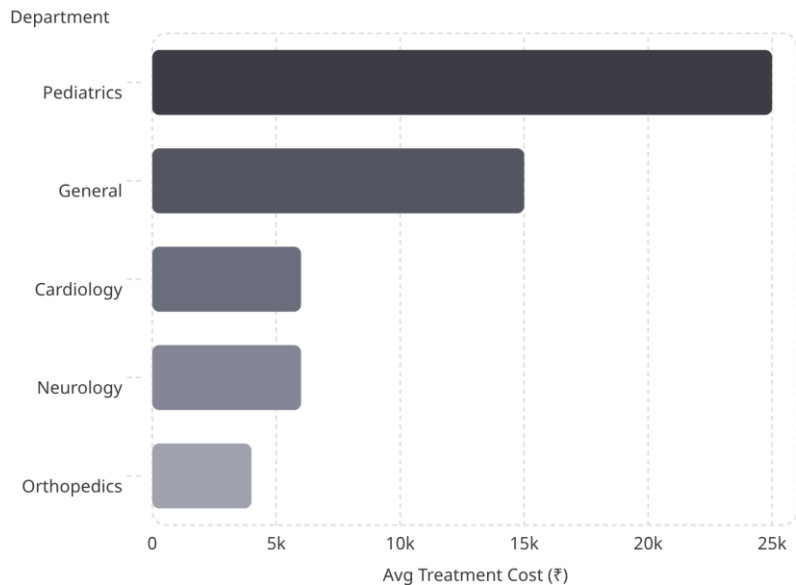
Uninsured Avg Bill

₹10,059

Insured Avg Bill

₹5,540

Departmental Cost Analysis



6× Cost Difference

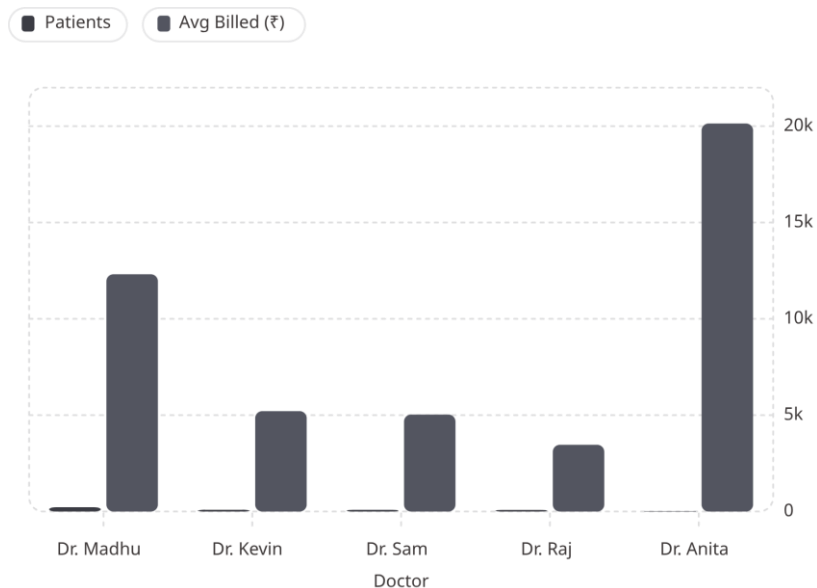
Pediatrics (₹25,000 avg) is **67% more expensive** than General and **525% more** than Orthopedics — yet sees only 18 patients (3.6%).

General handles **224 patients (44.8%)**, entirely Covid-19 driven — the largest revenue department by volume.



Commission a Pediatrics cost audit — ₹25,000 avg at only 18 patients needs justification.

Doctor Performance & Workload



Critical Imbalances

→ Dr. Madhu — 44.8% Caseload

224 patients, 10-day avg stay. Single point of failure — redistribute immediately.

→ Dr. Sam — Zero Stay Duration

87 patients, 0.1-day avg stay. Missing discharge records or data error.

→ Dr. Anita — Highest Per-Patient Cost

Only 18 patients but ₹20,139 avg billed. Audit for complexity or billing anomaly.

Patient Stay Duration Analysis

5.8

Avg Stay (Days)

64%

Long Stay Rate

320 of 500 patients stay >4 days

169

Elder Long Stays

65+ age group

10

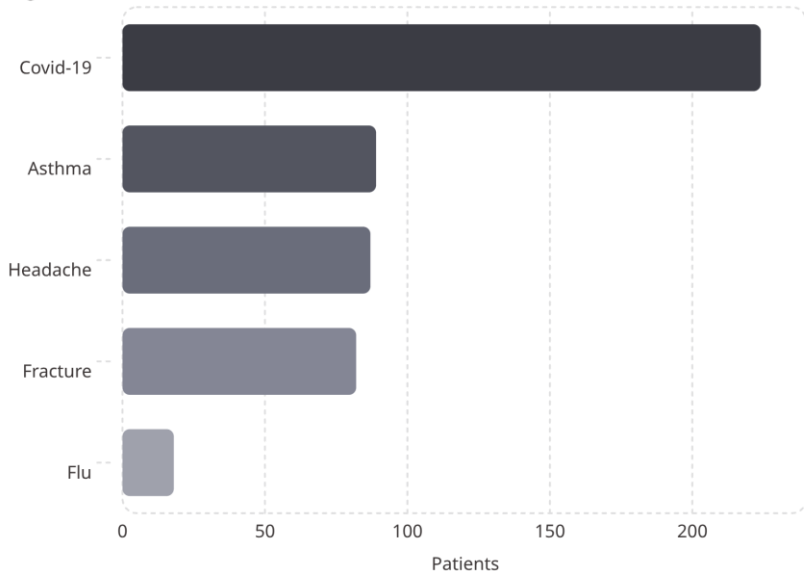
Max Stay

General department

Elder (65+) and Senior (46–60) patients account for **81% of all long stays**. General department avg stay reaches 10 days. Implement dedicated elder discharge planning protocols to free bed capacity.

Diagnosis & Patient Demographics

Diagnosis



Geographic Distribution

Covid-19 accounts for **44.8%** of all diagnoses — more than all other conditions combined. Each diagnosis maps 1:1 to a department.

Mumbai

175 patients · 35.0% — more than double the second city

Bengaluru · Hyderabad · Chennai · Delhi

91, 85, 82, 67 patients respectively

Gender Split

Near-equal across all cities — 52.4% Male, 47.6% Female

Data Quality & Methodology

636 data quality issues resolved in Excel Power Query prior to Power BI import. Cleaned dataset: 500 unique patient records.

Issue Type	Count	Resolution
Non-standard Gender	217	Replace Values → Male/Female
Non-standard Insurance	185	Replace Values → Yes/No
Negative Treatment Cost	110	Rows removed
Discharge before Admission	109	Dates corrected
Duplicate PatientIDs	10	Remove Duplicates
Negative Age Values	5	Rows removed

 Neurology and Dr. Sam avg stay of 0.1 days flagged — discharge dates may be missing for ~87 records.

Strategic Recommendations

01

Insurance Enrollment Program

Target Pediatrics and General departments. Partner with Ayushman Bharat to reduce the 82% cost gap.

02

Redistribute Caseload

Cap any single doctor at 30% of total patient volume. Dr. Madhu's 45% share is an operational risk.

03

Elder Discharge Protocols

Implement social care coordination in General department to reduce 10-day avg stays and free bed capacity.

04

Pediatrics Cost Audit

Investigate ₹25,000 avg per visit across only 18 patients. Benchmark against comparable facilities.

05

Data Verification

Audit Dr. Sam and Neurology records for missing discharge dates causing 0.1-day avg stay anomalies.